

Design Patterns Homework

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1. Suppose that you have several client objects that must interact with a database of models, each of which consists of various elements. The client must first open the database and query it to get a particular model. The model is then accessed to get an Element. The Element is then accessed to get the Element attributes. What design pattern is suggested?

Facade ✓

2. You are designing a graphics program that represents complex objects as an aggregation of simple geometric shapes: rectangle, circle, triangle, etc. A coworker has a graphical display package that draws all of the required shapes. However, the routine that draws a rectangle needs the coordinates of the four corners of the rectangle. Your application defines a rectangle in terms of the coordinates of the lower left corner and a height and width. What design pattern is suggested?

Adapter ✓

3. You are designing an application for a Chamber of Commerce that will be used to track Chamber members. When the data for a new member is added, you want to send a welcome letter to the new member. What design pattern is suggested?

Observer ✓

4. You are designing an objects database. The objects are composites of potentially both the fundamental data types: character, integer, double and any objects that have been defined. In other words, objects can contain either objects or fundamental data types. You want to treat all types equivalently. What design pattern is suggested?

Composite ✓

5. For the objects database that you are designing in 4 above, you want to be able to sequentially access all objects in the database, without regard to the type of the object. What design pattern is suggested?

Iterator ✓

6. You are designing a word processing program that will include drawing created by the application in 2 above. You are worried that opening a document with many figures will be extremely slow, because of the complex nature of the objects in the figures. You want the first page of the document to display as quickly as possible. What design pattern is suggested?

Proxy ✓

7. You want to add support for a new output device for a network of workstations. You must ensure that only one job is directed to the device at a time. What design pattern is suggested?

Singleton ✓